

Exercise 1

• Insert a new record in the customers table using the graphical tool offered by MySQL Workbench. The data must be the following:

Field Value

customerNumber 500

customerName Stellar Tech Ltd.

contactLastName Rivers

contactFirstName Julia

phone +34 600123456

addressLine1 99 Innovation Rd.

addressLine2 NULL

city Barcelona

state NULL

postalCode 08018

country Spain

salesRepEmployeeNumber 1056

creditLimit 75000.00

- Here we have already added the necessary data provided by the exercise.

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'classicmodels' database structure, with the 'customers' table selected. The 'Table: customers' pane shows the columns: customerNumber (PK, int(11)), customerName (varchar(50)), contactLastName (varchar(50)), contactFirstName (varchar(50)), phone (varchar(50)), addressLine1 (varchar(50)), addressLine2 (varchar(50)), city (varchar(50)), state (varchar(50)), postalCode (varchar(15)), country (varchar(50)), salesRepEmployeeNumber (int(11)), and creditLimit (decimal(10,2)).

The main window displays the 'customers' table with 1000 rows. The newly inserted record is highlighted in blue at the bottom of the list:

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
500	Stellar Tech Ltd.	Rivers	Julia	+34 600123456	99 Innovation Rd.		Barcelona		08018	Spain	1056	75000.00

The 'Output' pane at the bottom shows the execution of three SQL queries:

```
1 21:51:20 SELECT * FROM innography.book LIMIT 0, 1000
2 21:51:30 SELECT * FROM innography.book LIMIT 0, 1000
3 21:51:42 SELECT * FROM classicmodels.customers LIMIT 0, 1000
```

The output for the third query shows 122 rows returned.

- Having filled in the previous row, we click “Apply” and “Finish”

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

• Insert a new record into the employees table using the graphical tool. The data must be:

Field Value

employeeNumber 2001

lastName Ortega

firstName Marc

extension x201

email marc.o@classicmodels.com

officeCode 1

reportsTo 1002

jobTitle Sales Rep

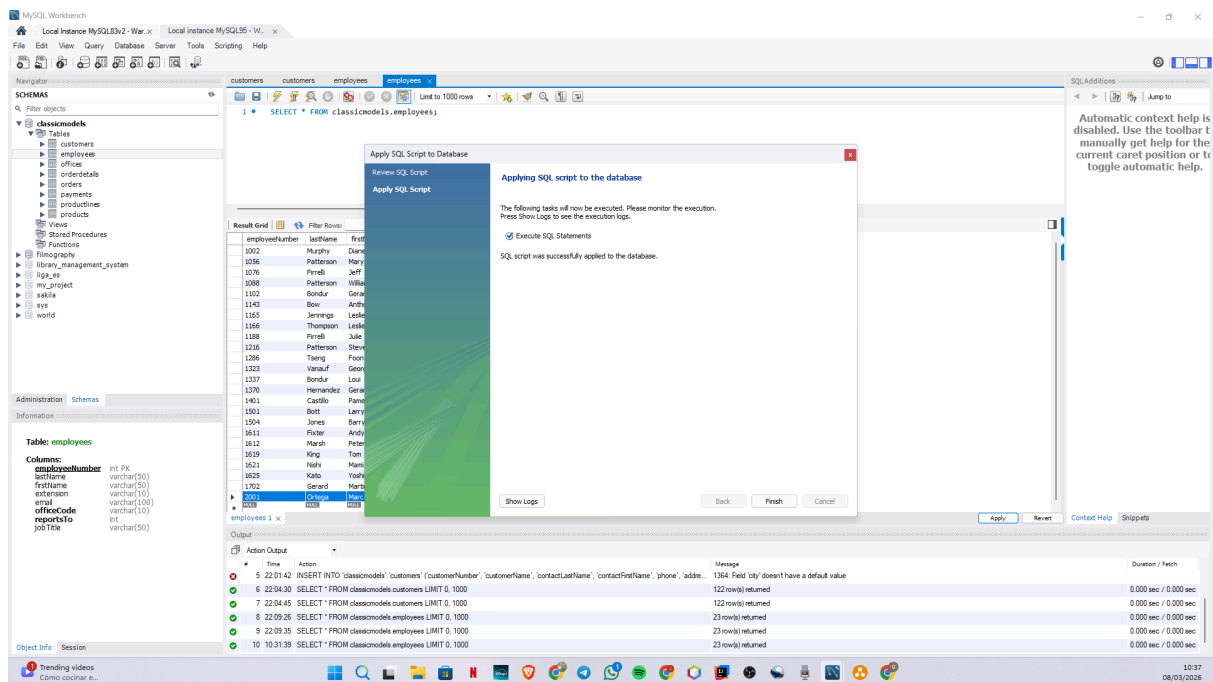
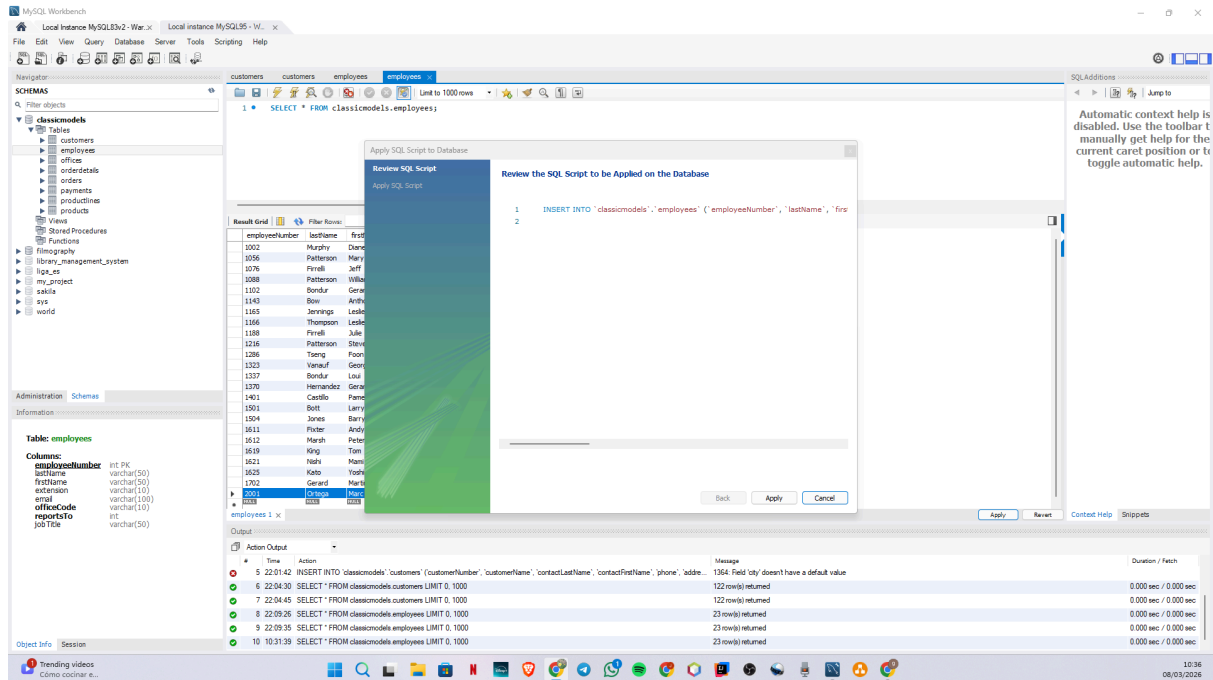
- As in the customers exercise, we added the line.

The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane displays the 'classicmodels' database with various tables and views. The 'employees' table is selected, and its structure is shown in the 'Table: employees' pane. The columns are: employeeNumber (INT PK), lastName (VARCHAR(50)), firstName (VARCHAR(50)), extension (VARCHAR(10)), email (VARCHAR(100)), officeCode (INT), reportsTo (INT), and jobTitle (VARCHAR(50)). The main pane shows a query result for the query: `SELECT * FROM classicmodels.employees;`. The result grid displays a list of employees, including the newly added record: `2001 Ortega Marc x201 marc.o@classicmodels.com 1 1002 Sales Rep`. The 'Output' pane shows the execution log, indicating that the query was successful and returned 23 rows.

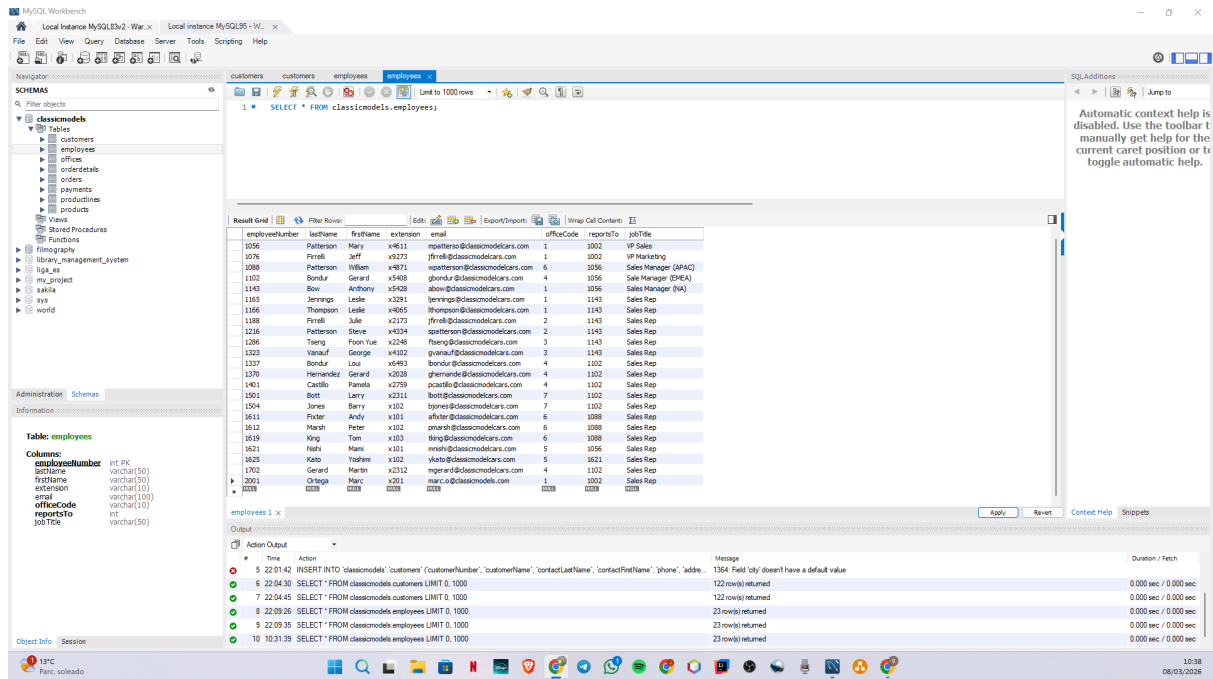
- Having already filled in the line with the information provided by the exercise.

This screenshot is similar to the previous one, but the new record (employeeNumber 2001) is highlighted in the result grid. The 'Output' pane shows the execution log, indicating that the query was successful and returned 23 rows. The new record is clearly visible in the result set, confirming its successful insertion into the database.

- Now we click on “Apply” > “Finish”.



- Line already added.



Exercise 2

Insert the following payments into the payments table using SQL statements:
customerNumber checkNumber paymentDate amount

124 H123 2024-02-06 845.00

151 H124 2024-02-07 70.00

112 H125 2024-02-05 1024.00

Copy the SQL statements you used in your submission. Only include the data shown.

- Statements

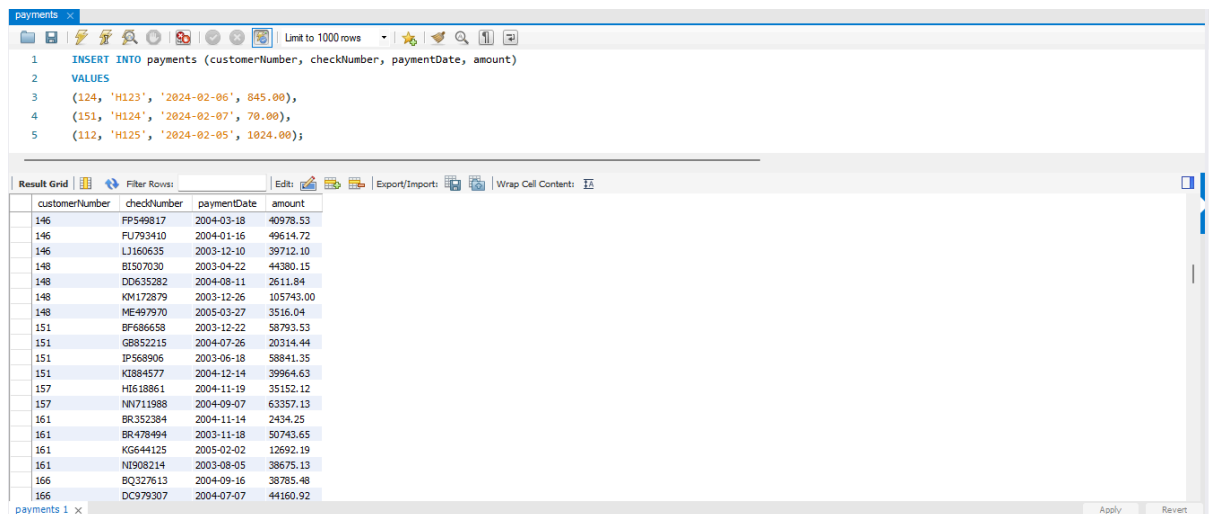
INSERT INTO payments (customerNumber, checkNumber, paymentDate, amount)

VALUES

(124, 'H123', '2024-02-06', 845.00),

(151, 'H124', '2024-02-07', 70.00),

(112, 'H125', '2024-02-05', 1024.00);



payments payments x

1 • SELECT * FROM classicmodels.payments;

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: 15

customerNumber	checkNumber	paymentDate	amount
145	ED39322	2004-04-26	28211.70
145	HR182688	2004-12-01	20564.86
145	JJ246391	2003-02-20	53959.21
146	FP549817	2004-03-18	40978.53
146	FU793410	2004-01-16	49614.72
146	LJ160635	2003-12-10	39712.10
148	BT507030	2003-04-22	44380.15
148	DD635282	2004-08-11	2611.84
148	KM172879	2003-12-26	105743.00
148	ME497970	2005-03-27	3516.04
151	BF686658	2003-12-22	58793.53
151	GB852215	2004-07-26	20314.44
151	H124	2024-02-07	70.00
151	IP568906	2003-06-18	58841.35
151	KI884577	2004-12-14	39964.63
157	HTE18861	2004-11-19	35152.12
157	NW711988	2004-09-07	63357.13
161	BR352384	2004-11-14	2494.25
161	BR478494	2003-11-18	50743.65

payments 1 x Apply Revert

Exercise 3

Modify the addresses of some customers using the graphical tool, and provide screenshots showing the changes made. Update the records in the customers table as follows:

customerNumber	city	addressLine1	postalCode
141	Girona	43 State St	17001
124	Lleida	2 Walls St	25002
119	Tarragona	46 Marlborough Rd.	43003

- Lines not modified.

customers 1 x

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: 15

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	Nantes	Nantes	FRANCE	44000	France	1370	21000.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	NV	Las Vegas	USA	83030	USA	1166	71800.00
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Australia	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labrun	Janine	40.67.8555	67, rue des Cinquante Otages	Nantes	Nantes	FRANCE	44000	France	1370	118200.00
121	Baane Mini Imports	Bergussen	Jonas	07-98 9555	Erling Skakkes gate 78	Norway	Stavern	Norway	4110	Norway	1504	81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551490	5677 Strong St.	CA	San Rafael	USA	97562	USA	1165	210500.00
125	Havel & Zbyszek Co.	Piestrzeniewicz	Zbyszek	(26) 642-7555	ul. Filtrowa 68	Poland	Warszawa	Poland	01-012	Poland	1504	0.00
128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonenstr. 34	Germany	Frankfurt	Germany	60528	Germany	1504	59700.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street	USA	San Francisco	USA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue	NY	NYC	USA	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	C/ Moralezal, 86	Spain	Madrid	Spain	28034	Spain	1370	227600.00
144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555	Berguvsvägen 8	Sweden	Luleå	Sweden	S-958 22	Sweden	1504	53100.00
145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555	Vinbæltet 34	Denmark	Kobenhavn	Denmark	1734	Denmark	1401	83400.00
146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555	2, rue du Commerce	France	Lyon	France	69004	France	1337	123900.00
148	Dragon Souvenirs, Ltd.	Natividad	Eric	+65 221 7555	Bronz Sok.	Singapore	Bronz Apt. 3...	Singapore	079903	Singapore	1621	103800.00
151	Muscle Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	NY	Suite 400	USA	10022	USA	1286	138500.00
157	Diecast Classics Inc.	Leong	Kevin	2155551555	7586 Pompton St.	PA	Allentown	USA	70267	USA	1216	100600.00
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle	CA	Burlingame	USA	94217	USA	1165	84600.00
166	Handy Gifts & Co.	Victorino	Wendy	+65 224 1555	106 Linden Road Sandown	Singapore	2nd Floor	Singapore	069045	Singapore	1612	97900.00

customers 1 x Apply Revert

- Code already modified but not applied.

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	Nantes	Nantes	FRANCE	44000	France	1370	21000.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	NV	Las Vegas	USA	83030	USA	1166	71800.00
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Australia	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labrun	Janine	40.67.8555	46 mMarlborough Rd.	Nantes	Tarragona	FRANCE	43003	France	1370	118200.00
121	Baane Mini Imports	Bergussen	Jonas	07-98 9555	Erling Skakkes gate 78	Norway	Stavern	Norway	4110	Norway	1504	81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551490	46 Walls St	CA	Lleida	USA	25002	USA	1165	210500.00
125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek	(26) 642-7555	ul. Filtrowa 68	Poland	Warszawa	Poland	01-012	Poland	1504	0.00
128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonenstr. 34	Germany	Frankfurt	Germany	60528	Germany	1504	59700.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street	USA	San Francisco	USA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue	NY	NYC	USA	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	43 State St	Spain	Girona	Spain	17001	Spain	1370	227600.00
144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555	Berguvsvägen 8	Sweden	Luleå	Sweden	S-958 22	Sweden	1504	53100.00
145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555	Vinbæltet 34	Denmark	Kobenhavn	Denmark	1734	Denmark	1401	83400.00
146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555	2, rue du Commerce	France	Lyon	France	69004	France	1337	123900.00
148	Dragon Souvenirs, Ltd.	Natividad	Eric	+65 221 7555	Bronz Sok.	Singapore	Bronz Apt. 3...	Singapore	079903	Singapore	1621	103800.00
151	Muscle Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	NY	Suite 400	USA	10022	USA	1286	138500.00
157	Diecast Classics Inc.	Leong	Kevin	2155551555	7586 Pompton St.	PA	Allentown	USA	70267	USA	1216	100600.00
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle	CA	Burlingame	USA	94217	USA	1165	84600.00
166	Handy Gifts & Co.	Victorino	Wendy	+65 224 1555	106 Linden Road Sandown	Singapore	2nd Floor	Singapore	069045	Singapore	1612	97900.00

- Applying.

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Review the SQL Script to be Applied on the Database

```

1  UPDATE `classicmodels`.`customers` SET `addressLine1` = '46 mMariborough Rd.'
2  UPDATE `classicmodels`.`customers` SET `addressLine1` = '46 Walls St', `city` =
3  UPDATE `classicmodels`.`customers` SET `addressLine1` = '43 State St', `city` =
4

```

Back

Apply

Cancel

- Everything has already been implemented.

Limit to 1000 rows

1

SELECT * FROM classicmodels.customers;

Result Grid

	customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine		40.32.2555	54, rue Royale	NULL	Nantes	NULL	44000	France	1370	21000.00
112	Signal Gift Stores	King	Jean		7025551838	8489 Strong St.	NULL	Las Vegas	NV	83030	USA	1166	71800.00
114	Australian Collectors, Co.	Ferguson	Peter		03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labruno	Janine		40.67.8555	46 mMariborough Rd.	NULL	Tarragona	NULL	43003	France	1370	118200.00
121	Baane Mini Imports	Bergulfien	Jonas		07-98 9555	Erling Skakkes gate 78	NULL	Stavern	NULL	4110	Norway	1504	81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan		4155551490	46 Walls St	NULL	Lleida	CA	25002	USA	1165	210500.00
125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek		(26) 642-7555	ul. Filbrowa 68	NULL	Warszawa	NULL	01-012	Poland	NULL	0.00
128	Blauer See Auto, Co.	Keitel	Roland		+49 69 66 90 2555	Lyonerstr. 34	NULL	Frankfurt	NULL	60528	Germany	1504	59700.00
129	Mini Wheels Co.	Murphy	Julie		6505555787	5557 North Pendale Street	NULL	San Francisco	CA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai		2125557818	897 Long Airport Avenue	NULL	NYC	NY	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego		(91) 555 94 44	43 State St	NULL	Grona	NULL	17001	Spain	1370	227600.00
144	Volvo Model Replicas, Co	Berglund	Christina		0921-12 3555	Berguvsvägen 8	NULL	Luleå	NULL	S-958 22	Sweden	1504	53100.00
145	Danish Wholesale Imports	Petersen	Jytte		31 12 3555	Vinbæltet 34	NULL	Kobenhavn	NULL	1734	Denmark	1401	83400.00
146	Saveley & Henriot, Co.	Saveley	Mary		78.32.5555	2, rue du Commerce	NULL	Lyon	NULL	69004	France	1337	123900.00
148	Dragon Souvenirs, Ltd.	Natividad	Eric		+65 221 7555	Bronz Sok.	Bronz Apt. 3...	Singapore	NULL	079903	Singapore	1621	103800.00
151	Muscle Machine Inc.	Young	Jeff		2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA	1286	138500.00
157	Diecast Classics Inc.	Leong	Kelvin		2155551555	7586 Pompton St.	NULL	Allentown	PA	70267	USA	1216	100600.00
161	Technics Stores Inc.	Hashimoto	Juri		6505556809	9408 Furth Circle	NULL	Burlingame	CA	94217	USA	1165	84600.00
166	Handy Gifts Co	Victorino	Wendy		+65 224 1555	106 Linden Road Sandown	2nd Floor	Singapore	NULL	069045	Singapore	1612	97900.00

Exercise 4

Cancel the order made on 2003-09-28, using SQL instructions. Change the status to 'Cancelled', the shippedDate to the current date, and comments to 'Order cancelled due to delay'. Do this using a single SQL statement.

Copy the statement in your submission.

- Statement.

```
UPDATE orders
SET status = 'Cancelled',
    shippedDate = CURDATE(),
    comments = 'Order cancelled due to delay'
WHERE orderDate = '2003-09-28';
```

Exercise 5

Update all product names of type Trains to include the product code in parentheses. For

example, a product with productCode = S10_1949 and productName = "Vintage Train" will become:

"Vintage Train (code S10_1949)"

Do this with a single SQL statement using the proper MySQL functions.

Resource: MySQL functions - W3Schools

```
SELECT * FROM products WHERE productLine = 'Trains';
```

```
UPDATE products
SET productName = CONCAT(productName, ' (code ', productCode, ')')
WHERE productLine = 'Trains';
```

Exercise 6

Increase buyPrice and MSRP of all products with quantityInStock > 500 by 0.02%.

Do this with one SQL statement.

```
UPDATE products
SET buyPrice = buyPrice * 1.0002,
    MSRP = MSRP * 1.0002
WHERE quantityInStock > 500;
```

Exercise 7

Remove all payments made by customers who are represented by employees with the last name 'Patterson'.

Use a single SQL statement and copy it for the task delivery.

```
1. Find employees in Patterson
SELECT employeeNumber
FROM employees
WHERE lastName='Patterson';
```

2. Customers of those employees


```
SELECT customerNumber
FROM customers
WHERE salesRepEmployeeNumber IN (...)
```

3. Delete payments

```
DELETE FROM payments
WHERE customerNumber IN (...)
```

4. Final Consultation

```
DELETE FROM payments
WHERE customerNumber IN (
  SELECT customerNumber
  FROM customers
  WHERE salesRepEmployeeNumber IN (
    SELECT employeeNumber
    FROM employees
    WHERE lastName = 'Patterson'
  )
);
```

Exercise 8

**Delete all customers from Lisbon who have not made any payment.
Use one SQL statement only.**

1. Lisbon Clients

```
SELECT *
FROM customers
WHERE city='Lisbon';
```

2. Payment customers

```
SELECT customerNumber
FROM payments;
```

3. Exclude those customers

```
NOT IN
```

4. Final Consultation

```
DELETE FROM customers
WHERE city = 'Lisbon'
AND customerNumber NOT IN (
  SELECT customerNumber
  FROM payments
);
```

Exercise 9

Add all customers as new employees, using their contact names as first and last name.

Use customerNumber + 2000 as the new employeeNumber, leave all other fields as 'x0000' extension, 'new@company.com' email, '1' officeCode, 'Sales Rep' jobTitle

Use contactFirstName as firstName, contactLastName as lastName.

Do this with a single SQL statement.

```
INSERT INTO employees (  
  employeeNumber,  
  lastName,  
  firstName,  
  extension,  
  email,  
  officeCode,  
  jobTitle  
)  
SELECT  
  customerNumber + 2000,  
  contactLastName,  
  contactFirstName,  
  'x0000',  
  'new@company.com',  
  '1',  
  'Sales Rep'  
FROM customers;
```

Exercise 10

Cancel all orders made by customers handled by the customer Elizabeth Lincoln.

Change the status to 'Cancelled', shippedDate to the current date, and comments to 'Order cancelled by management'.

Use a single SQL statement.

```
1. Encontrar cliente  
SELECT customerNumber  
FROM customers  
WHERE contactFirstName='Elizabeth'  
AND contactLastName='Lincoln';
```

```
2. Cancelar pedidos  
UPDATE orders  
SET status='Cancelled'  
WHERE customerNumber IN (...)
```

3. Consulta final

```
UPDATE orders
SET status = 'Cancelled',
    shippedDate = CURDATE(),
    comments = 'Order cancelled by management'
WHERE customerNumber IN (
    SELECT customerNumber
    FROM customers
    WHERE contactFirstName = 'Elizabeth'
    AND contactLastName = 'Lincoln'
);
```